

Causal Strength Under Structure Uncertainty

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Abstract

How should a learner estimate the strength of a causal influence when it is uncertain whether the causal relation exists? We address this question using the framework of Bayesian causal induction. Existing accounts have primarily focused either on inference about causal structure or on estimation of causal strength under an assumed structure. We propose *structure-averaged causal strength* as a method for estimating causal strength under structure uncertainty. By averaging causal strength across alternative structures according to their posterior probabilities, the account provides a strength estimate that reflects uncertainty about both whether a causal relation exists and how strong it is. Normatively, our analyses clarify how causal strength should be estimated when the existence of the causal relation itself remains uncertain. Empirically, structure-averaged strength provides accurate predictions of human judgments of causal influence across a broad set of published studies, suggesting that human causal judgments reflect uncertainty about both causal strength and causal structure. Together, our analyses show how inferences about causal structure shape judgments of causal strength under uncertainty.

Keywords: causal induction, causal structure, causal judgment, causal strength, Bayesian inference, computational modeling, causal learning

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Introduction

Causal learning is fundamental to human cognition and behavior. Observing that an outcome follows an action or that recovery follows a treatment can provide evidence for a causal relation, but neither the existence nor the strength of that relation can be observed directly (Waldmann, 2017). Causal induction therefore requires learners to draw inferences about latent causal relations from limited and noisy evidence.

From a computational perspective, causal induction involves two related but distinct problems: inferring whether a causal relation exists and estimating how strong that relation is, if it exists. These correspond to inference about causal structure and estimation of causal strength (Griffiths & Tenenbaum, 2005). Consider an initial clinical trial in which patients in the intervention group recover more often than patients in the control group. The observed difference may suggest a causal effect of the treatment, but with limited evidence it remains uncertain both whether the treatment causes recovery and how strong a potential effect is. This raises a central question that is the focus of the present work: What does it mean to estimate causal strength when the existence of the causal relation itself is uncertain?

Bayesian models of causal induction provide a computational framework for addressing this question because they represent uncertainty over both causal structures and their parameters (Griffiths & Tenenbaum, 2005, 2009; Lu et al., 2008; Meder et al., 2014; Stephan & Waldmann, 2018). On this view, learners consider alternative hypotheses about the underlying causal relations and use the observed data to update beliefs about both which structure generated the observations and how strong the involved causal relations are. This provides a formal framework for psychological theories according to which human learning and reasoning reflect inferences about latent causal models rather than observed covariation (Danks, 2014; Gopnik et al., 2004; Sloman & Hagmayer, 2006; Waldmann, 1996; Waldmann et al., 2006).

The distinction between structure and strength has direct implications for the normative and descriptive analysis of human causal judgments. In a typical causal-learning task, participants observe a sample of cases in which a candidate cause and an outcome are either present or absent, and then rate the magnitude of the causal relation (e.g., on a scale from 0 to 100). These tasks have typically been designed to examine how people assess the strength of a relation, with formal measures such as contingency (Cheng & Novick, 1992; Jenkins & Ward, 1965) and causal power (Cheng, 1997) serving as normative benchmarks. However, a causal rating may reflect not only the estimated magnitude of a relation, but also the learner’s beliefs about how likely a link exists given the observed evidence. Consistent with this ambiguity, structural quantities have been shown to predict judgments elicited by questions that nominally ask about causal strength (Griffiths & Tenenbaum, 2005, 2009). Thus, task wording alone does not determine whether a causal rating should be interpreted as a judgment about link existence, causal magnitude, or both.

Although the Bayesian framework represents uncertainty at both the level of causal structure and causal strength, existing Bayesian accounts of human causal judgments tend to focus on one level of inference at a time. Structural quantities indicate how strongly the data favor the presence of a causal link or how probable such a link is (Griffiths & Tenenbaum, 2005, 2009). Strength-related quantities in turn estimate the magnitude of a causal influence under an assumed structure (Lu et al., 2008; Yeung & Griffiths, 2015).

We argue that both uncertainty about structure and uncertainty about causal strength are relevant when estimating the influence of a candidate cause and propose *structure-averaged causal strength*, which integrates inference about whether a causal link exists with estimation of how strong the link is if it exists.

We make three contributions. First, we formalize structure-averaged strength using Bayesian model averaging and derive model predictions across different learning environments. Second, we show how structure-averaged strength differs both from judgments about the existence of a link and from Bayesian strength estimates under a single causal structure. This clarifies which quantities are normatively appropriate for different inferential questions and why the same observations support different kinds of causal judgments. Third, we evaluate the descriptive relevance of structure-averaged strength by revisiting a broad set of published studies on human causal judgment. Our findings suggest that participants’ judgments reflect both estimated causal magnitude and uncertainty about whether a causal relation exists. Together, these analyses clarify what it means to estimate causal strength under structure uncertainty.

From Contingency to Causal Strength

In standard causal-learning tasks, learners observe whether a binary candidate cause C (e.g., a virus) and effect E (e.g., a disease) are present or absent across a series of cases. Descriptively, the observed sample can be summarized by the conditional probability of the effect when the cause is present, $P(e | c)$, and the conditional probability of the effect when the cause is absent, $P(e | \neg c)$. Their difference defines the observed contingency between the two events:

$$\Delta P = P(e | c) - P(e | \neg c). \quad (1)$$

Positive values of $\Delta P > 0$ indicate that the effect occurs more frequently when the candidate cause is present, whereas negative values indicate that the effect occurs less frequently in its presence ($\Delta P < 0$). Thus, the sign of ΔP captures whether the candidate cause raises or lowers the probability of the effect, and its magnitude captures the size of that change. When the effect is equally likely in the presence and absence of the candidate cause, the observed contingency is zero, $\Delta P = 0$.

In theories of causal learning, ΔP can be interpreted as an estimate of causal strength under a linear-additive model, in which the candidate cause changes the probability of the effect by a constant amount (Cheng & Novick, 1992; Waldmann & Holyoak, 1992). More broadly, this notion of probabilistic “difference making” has played a central role in research on learning, causation, and information search (Danks, 2003; Jenkins & Ward, 1965; Nelson et al., 2022; Sprenger, 2018; Tentori et al., 2007).

Interpreting ΔP as a measure of causal strength corresponds to assuming that the cause has an additive influence on the probability of the effect (Griffiths & Tenenbaum, 2005; Meder et al., 2025). A different measure results when the candidate cause and background causes are assumed to provide independent opportunities for producing the effect, a functional form known as noisy-OR gate (Glymour, 2003; Pearl, 1988). In this case, *causal power* (Cheng, 1997), denoted q_c , represents the corresponding measure of causal strength. Causal power is defined as

$$q_c = \frac{\Delta P}{1 - P(e \mid \neg c)}. \quad (2)$$

This estimate adjusts the observed contingency for the proportion of cases in which the effect was not already produced by background causes and therefore was available to be produced by the candidate cause. For preventive relations, the corresponding formulation is based on a noisy-AND-NOT model (Griffiths & Tenenbaum, 2005; Yuille & Lu, 2007), in which the candidate cause probabilistically prevents an effect that would otherwise be produced by background causes.

Both models, ΔP and causal power, thus estimate the strength of a causal influence, but under different assumptions about how the candidate cause and background causes combine: ΔP corresponds to a linear-additive representation, whereas causal power follows from noisy-OR or noisy-AND-NOT assumptions. What they have in common is that both provide point estimates of causal strength under an assumed causal structure containing a link between C and E . Although zero causal strength is included as a possible parameter value, the possibility that no link exists at all is not represented as a distinct hypothesis whose plausibility can be evaluated.

Two fundamental sources of uncertainty are therefore not taken into account. First, a learner faces uncertainty about the magnitude of the causal influence. Due to sampling variability, a weak causal influence may appear strong in a particular sample, or a strong influence may produce little observed contingency by chance. Thus, even under a given causal structure, the observed data may be compatible with a range of underlying strength values. Second, a learner faces uncertainty about whether the assumed causal relation exists at all. The same observations may be compatible with alternative causal structures: an observed association may have been generated by a causal link, but may also have arisen in the absence of such a link. Causal induction under uncertainty therefore requires representing both *parameter uncertainty* about the magnitude of a causal influence and *structure uncertainty* about whether a causal link exists. Bayesian causal induction provides a framework for capturing both forms of uncertainty.

Existing Bayesian Models of Causal Structure and Strength

Bayesian causal induction distinguishes inference about causal structure from inference about parameters within a structure (Griffiths & Tenenbaum, 2005, 2009; Lu et al., 2008; Meder et al., 2014; Stephan & Waldmann, 2018). Psychologically, this framework aligns with the idea that people reason at the causal level and thereby go “beyond the information given” (Waldmann et al., 2006). On this view, learners do not merely assess observed covariation or estimate the strength of an assumed relation, but interpret the observations as evidence for latent causal relations. Existing Bayesian accounts of human causal judgment have used this framework to address two central questions of causal inference under uncertainty: whether a causal link exists and how strong that link is, assuming it exists.

Inference About Causal Structure

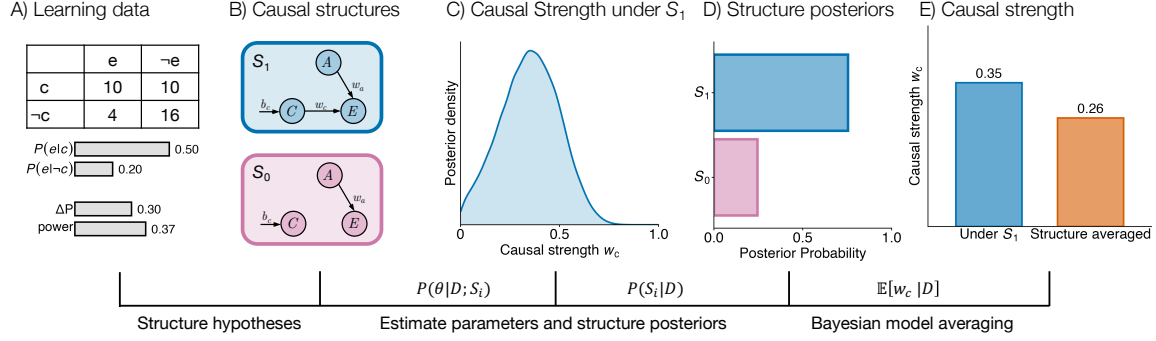
Within the framework of Bayesian causal induction, learning is modeled as inference over graphical causal models, with uncertainty represented at both the parameter and structure levels. The learner considers alternative hypotheses about the data-generating process and infers their posterior probabilities given the observed data. In a simple causal-learning task, the joint occurrence of C and E can be summarized in a 2×2 contingency table (Figure 1A). These observations can then be evaluated under two alternative causal structures (Figure 1B). Structure S_1 represents the hypothesis that candidate cause C is causally related to effect E . Under S_1 , the probability of E depends on both C and unobserved background causes A , with w_c denoting the strength of C and w_a the aggregate influence of A . Structure S_0 , by contrast, represents the hypothesis that C has no causal influence on E , such that the effect is generated solely by background causes. Under S_0 , any observed association between C and E is incidental and does not reflect a causal influence.

Within each structure, uncertainty about parameters such as w_c and w_a (strength of candidate and background cause, respectively) is represented by probability distributions rather than single point estimates. Figure 1C illustrates the resulting posterior distribution over the causal strength parameter w_c under S_1 , derived here from uniform parameter priors and the assumption that the candidate cause and background causes provide independent opportunities to produce the effect (i.e., a noisy-OR parameterization). Its posterior mean provides a Bayesian estimate of causal strength under the link model S_1 , where causal power corresponds to the maximum-likelihood estimate of w_c under a noisy-OR model (Griffiths & Tenenbaum, 2005). The interpretation of w_c thus depends on the assumed functional form; in the noisy-OR and noisy-AND-NOT models considered here, it corresponds to generative or preventive causal power, respectively (Cheng, 1997).

Evaluating the competing structures requires averaging each structure’s likelihood, $P(D|S_i)$, over its possible parameter values. This yields a marginal likelihood for each structure, reflecting how well it predicts the observations after accounting for parameter uncertainty. Because S_0 and S_1 define distinct hypotheses and likelihood functions, the simpler no-link structure may be favored when it predicts the observations better than the more complex link structure, consistent with Bayesian Occam’s razor (MacKay, 2003). The marginal likelihoods are then combined with prior probabilities over structures to obtain posterior structure probabilities, $P(S_i|D)$. Figure 1D illustrates the posterior probabilities assigned to S_1 and S_0 , assuming equal prior probabilities over the two structures. These posterior structure probabilities quantify uncertainty about which causal model generated the observations and provide the basis for measures that capture inference about causal structure. Thus, unlike contingency and causal power, posterior structure probabilities quantify uncertainty about whether the observed data were generated by a causal relation at all.

Posterior Link Probability as Structural Evidence

A fundamental challenge for learners is to discover the causal structure of their environment. In the terms of Tolman and Brunswik (1935), organisms inhabit a complex “causal texture” in which events are linked through causal relations, and learning about individual relations provides information about this broader causal organization.

**Figure 1**

Bayesian causal induction and structure-averaged strength. A) Observed learning data, summarized by the joint frequency distribution of cause and effect, together with the corresponding conditional probabilities and sample estimates of ΔP and causal power. B) Alternative causal structures. Under S_1 , the candidate cause C and background causes A influence the effect E , with strengths w_c and w_a , respectively. Under S_0 , only the background causes influence the effect, and the causal strength of C is fixed at zero. Parameter b_c denotes the probability that the candidate cause is present. C) Posterior distribution over causal strength under the link structure, $P(w_c | D, S_1)$, derived under independent uniform priors over all parameters. D) Posterior probabilities of the link and no-link structures, $P(S_1 | D)$ and $P(S_0 | D)$, given equal prior probability. E) Causal strength under S_1 and structure-averaged causal strength, obtained by weighting the strength implied by each structure by its posterior probability.

In its simplest form, this problem involves determining whether a particular causal relation exists. Within the Bayesian framework, the comparison between link and no-link hypotheses gives rise to several structure-related measures of causal evidence. On this view, learning changes not only an estimate of causal strength but also beliefs about causal structure. Griffiths and Tenenbaum (2005) initially formalized structural evidence in terms of *causal support*, defined as the log Bayes factor (Kass & Raftery, 1995) comparing the marginal likelihood of the data under the link and no-link structure. Causal support therefore quantifies how strongly the observed data favor S_1 over S_0 . Griffiths and Tenenbaum (2009) subsequently focused on the posterior probability of the link structure, $P(S_1 | D)$, which captures how likely it is that a causal relation exists, given the data and prior beliefs in the presence of a link. $P(S_1 | D)$ can be obtained from causal support by combining the Bayes factor with the prior odds of the link structure (Griffiths & Tenenbaum, 2009).

Posterior link probability answers a different question from ΔP or causal power, which evaluate the strength of a causal relation and hence are methods for parameter estimation. Whereas these measures estimate how strong a causal influence is under an assumed link structure, $P(S_1 | D)$ represents the learner’s uncertainty about whether the link exists at all. Notably, Griffiths and Tenenbaum (2009) found that posterior link probability correlated strongly with causal judgments elicited using wording about the degree of causal influence. This finding suggests that uncertainty about causal structure may be reflected even in judgments commonly interpreted as estimates of causal strength, motivating

posterior link probability as a candidate descriptive model of such judgments.

Bayesian Causal Strength Under Sparse-and-Strong Priors

Bayesian accounts of causal induction extend traditional strength measures such as contingency and causal power by providing inference mechanisms for questions about causal structure. In typical causal-learning tasks, however, learners are asked to judge how strong a relation is, raising the question of how the same framework can account for uncertainty when estimating the causal magnitude of a presumed relation. Lu et al. (2008) developed a Bayesian version of causal power in which the strength of the candidate cause and influence of the background cause are treated as uncertain parameters. We focus on the sparse-and-strong prior formulation because it provides a psychologically motivated Bayesian account of causal strength judgment. Under this formulation, learners infer the strength under the assumption of the link structure S_1 , consistent with the assumption underlying causal power that the candidate causal relation is present (Cheng, 1997; Glymour, 2003). However, rather than deriving a single estimate directly from the observed frequencies, the model combines the likelihood of the data with prior distributions over these parameters. The resulting posterior distribution represents the strength of C that remain plausible after observing the data, and its posterior mean provides a Bayesian estimate of causal strength conditional on the presence of the link.

Lu et al. (2008) emphasized the role of prior expectations in causal learning, proposing that learners are guided by generic principles such as causal simplicity. They formalized this idea through sparse-and-strong (SS) priors, which favor causal systems with relatively few effective causes, each tending to be strong. For generative relations, the prior couples w_c and w_a , favoring either a strong candidate cause with weak background causes or the reverse. As a result, posterior strength estimates can differ from sample causal power and from estimates obtained under uniform parameter priors or when prior expectations are estimated empirically (Yeung & Griffiths, 2015). Psychologically, the SS prior formalizes an inductive bias toward sparse causal systems with strong causes. The resulting estimate represents uncertainty about causal magnitude inferred under the default structure S_1 , but does not capture structural uncertainty about whether the link exists.¹

Causal Strength Under Structure Uncertainty

Posterior link probability and Bayesian strength under the link structure answer different questions about an uncertain causal relation. Posterior link probability represents uncertainty about how likely it is that a causal relation exists, whereas Bayesian strength represents how strong that relation is, assuming the link model S_1 as default structure. Neither quantity, however, provides an estimate of causal strength that incorporates structural uncertainty about whether the link exists. This leaves a central question open, which the present work addresses: How should causal strength be estimated when there is uncertainty whether the link indeed exists? This question is normatively relevant because present accounts do not address how structure uncertainty can be incorporated into rational strength

¹Lu et al. (2008) also modeled structure inference by comparing link and no-link structures under an augmented SS prior. The present analyses use the basic SS prior and focus on the posterior mean of w_c under S_1 .

estimation. It is also psychologically relevant: the finding that posterior link probability can account for judgments elicited by questions about the magnitude of a causal influence suggests that uncertainty about the presence of a link matters even when learners are asked to judge causal strength.

To illustrate, suppose a learner observes the data shown in Figure 1A. The effect occurs more frequently when the candidate cause is present than when it is absent. This observation provides evidence for a generative causal relation whose strength one could describe in terms of ΔP or causal power. Yet with only 40 observations, a sample size typical of many causal-learning experiments, it remains plausible that the observed difference does not reflect a genuine causal influence. Estimating causal strength under the link model S_1 therefore describes how strong the cause would be *if* a link indeed existed, but does not take into account the possibility that there is no causal relation at all.

We address this problem by defining causal strength as a posterior expectation across alternative causal structures. We refer to this quantity as *structure-averaged causal strength*. An early account based on this idea was proposed to model trial-by-trial learning curves (i.e., dynamic learning; Danks et al., 2002), and analogous averaging principles have been applied to diagnostic inference (Meder et al., 2014) and causal attribution (Stephan & Waldmann, 2018). The present account uses this principle to address the problem of estimating causal strength under structure uncertainty and examines its normative and descriptive implications relative to existing structure- and strength-based accounts. Normatively, the model provides an estimate of causal strength that incorporates uncertainty about both the magnitude and the existence of the causal relation. Psychologically, it provides an account of strength judgments in which uncertainty about whether a causal relation exists influences the estimated magnitude of that relation.

Structure-Averaged Causal Strength

To estimate causal strength while retaining structural uncertainty, we use Bayesian model averaging (Hinne et al., 2020; Hoeting et al., 1999). Conceptually, the posterior mean of causal strength within each structure is weighted by that structure’s posterior probability. Formally, it is defined as

$$\mathbb{E}[w_c \mid D] = \sum_{S_i \in \mathcal{S}} \mathbb{E}[w_c \mid D, S_i] \cdot P(S_i \mid D), \quad (3)$$

where $\mathbb{E}[w_c \mid D, S_i]$ is the posterior mean strength under each considered structure, and $P(S_i \mid D)$ denotes the structure’s posterior probability. This account incorporates parameter uncertainty through the posterior distribution over causal strength within each structure and captures structure uncertainty through the posterior probabilities assigned to the alternative structures.

Figure 1 illustrates this computation, assuming a noisy-OR generating function, uniform priors over the model parameters, and equal prior probabilities for structures S_0 and S_1 . Panels A and B show the observed data and the link and no-link structures. Panel C shows the posterior distribution of causal strength under S_1 , and Panel D shows the posterior probabilities of the two structures. Because the data, due to the limited sample size, do not conclusively establish that a causal relation exists, the posterior probability of

S_0 remains substantially greater than zero. Structure-averaged causal strength combines structural evidence with the strength estimate by weighting the strength implied by each structure by its posterior probability (Panel E). Because S_0 contributes zero strength while retaining substantial posterior probability, the structure-averaged estimate is lower than the estimate under S_1 . Accordingly, structure-averaged strength is high only when both the strength estimate under S_1 and the posterior probability of S_1 are high.

In the case considered here, the no-link structure fixes the causal strength of C at zero. Structure-averaged strength therefore reduces to the posterior mean under S_1 weighted by the posterior probability of S_1 . Accordingly, the strength estimate under S_1 is attenuated whenever there is remaining uncertainty about the existence of a causal link. However, the general principle can be applied to any number of alternative causal structures, with the strength implied by each structure weighted by its posterior probability. For example, when the causal framing allows for both generative and preventive relations, we can average causal strength across generative, preventive, and no-link structures (cf. Danks et al., 2002, and analyses below). The same principle extends to more complex sets of causal structure hypotheses and other forms of causal inference (Meder et al., 2014; Stephan & Waldmann, 2018), although practical application to large hypothesis spaces may require approximations to exhaustive structure averaging.

From a normative perspective, structure-averaged strength thus addresses the question of how to estimate causal strength when it remains uncertain whether a causal link exists. This quantity reflects both uncertainty about the magnitude of a causal relation and structural uncertainty about whether that relation exists at all. Computationally, structure averaging also clarifies the relation to existing accounts. When the evidence strongly supports the link structure and the posterior probability of S_1 approaches one, structure-averaged strength converges to the strength estimate under S_1 . Accordingly, in the two-structure case considered here the causal strength estimate under S_1 represents the upper bound of structure-averaged causal strength. Conversely, the lower the probability of a link given the data, the more structure-averaged strength is pulled towards zero. Psychologically, this implies that judgments of causal strength should depend not only on the strength inferred under a particular causal structure, but also on the learner’s uncertainty about whether that structure is correct. The averaging principle itself is independent of the particular parameter priors used within each structure, although its specific predictions depend on both the parameter priors and the prior probabilities assigned to the alternative structures (Stephan & Waldmann, 2026).

At the level of an individual learner, structure-averaged strength represents expected causal strength under epistemic uncertainty about whether a causal link exists. The same averaging principle also offers an interpretation in terms of population frequencies. Consider a treatment that acts by binding to a particular receptor on tumor cells. If only some patients express this receptor, the causal mechanism through which the drug operates is present in some patients but absent in others. The expected causal effect across the population is then obtained by weighting the strength of the treatment when the mechanism is present by the prevalence of that mechanism in the population. Population heterogeneity can in turn give rise to epistemic uncertainty at the individual level: if it is unknown whether a particular patient’s tumor expresses the receptor, its prevalence in the relevant population provides information about the probability that the causal mechanism is present

in that patient. This interpretation also connects structure uncertainty to psychological theories emphasizing the role of causal mechanisms in human causal reasoning (e.g., Johnson & Ahn, 2017).

More generally, structure-averaged strength provides a psychological account of causal judgments in which uncertainty about whether a relation exists influences a learners estimates of causal strength. When the evidence strongly supports a causal link, strength judgments should primarily reflect the inferred magnitude of the relation. However, when the observed data leaves substantial uncertainty about whether a link exists, strength judgments should be attenuated relative to estimates of causal strength derived under the link structure. This approach also provides an alternative explanation for why structural quantities such as posterior link probability correlate with judgments elicited by questions about causal strength: human causal judgments may be sensitive to structure uncertainty without being judgments of structure alone.

Predictions of Structure-Averaged Causal Strength

Structure-averaged strength, posterior link probability, and Bayesian causal power represent distinct quantities that capture different aspects of causal induction. Although these quantities can produce similar qualitative patterns, they often differ quantitatively, for instance because evidence for a link and evidence about its strength need not coincide. We first illustrate these divergences across positive, zero, and negative contingencies. We then focus on causal induction under zero contingency, which has played a central role in advancing Bayesian accounts because positive causal judgments in the absence of an observed contingency can arise from rational inference under uncertainty even when contingency-based measures equal zero, hence implying the absence of causality.

Model Predictions Across Varying Contingencies

Figure 2 illustrates how structure-averaged strength makes distinct predictions across prototypical positive, zero, and negative contingencies. The positive and negative contingencies are evaluated under noisy-OR (generative) and noisy-AND-NOT (preventive) link structures, respectively. For the zero-contingency example, the hypothesis space contains a noisy-OR link structure and a no-link structure. Posterior link probability and structure-averaged strength were computed with equal prior probabilities over the competing structures and uniform priors over their parameters, whereas Bayesian strength is estimated under the link structure using SS priors. Estimates for preventive relations are shown as positive magnitudes to facilitate comparison of the estimated strength of the relation across conditions, irrespective of causal direction.

Across the three contingency conditions, all accounts predict the same qualitative ordering of causal judgments: highest for the positive contingency, lowest for the zero contingency, and intermediate for the negative contingency. However, the predictions differ substantially in magnitude, with posterior link probability yielding the highest estimates and structure-averaged strength the lowest. To illustrate the source of these differences, consider the positive-contingency example in the upper panel of Figure 2. Posterior link probability estimates how likely a causal relation exists and therefore yields a high value even though the observed contingency is moderate. Here, the posterior probability that a causal

link is close to certainty, although the strength of the link is only moderate. Bayesian strength estimation instead asks how strong the relation is conditional on its existence, yielding a more moderate estimate of causal strength. Structure-averaged strength differs from both accounts because it additionally reflects uncertainty about whether the relation exists at all, reducing the expected causal strength further.

Thus, the three approaches reflect different inferences from the same observations: evidence that a causal link exists, its expected strength conditional on that link, and its expected strength when uncertainty about the link itself is taken into account. Although the accounts yield similar qualitative patterns, they provide quantitatively distinct predictions across different causal environments. Notably, all accounts yield positive estimates under zero contingency, illustrating that an observed contingency of zero need not imply an inference of zero causal influence under uncertainty. Because the consequences of the different inferential mechanisms become especially apparent under zero-contingency learning, we examine this condition more closely next.

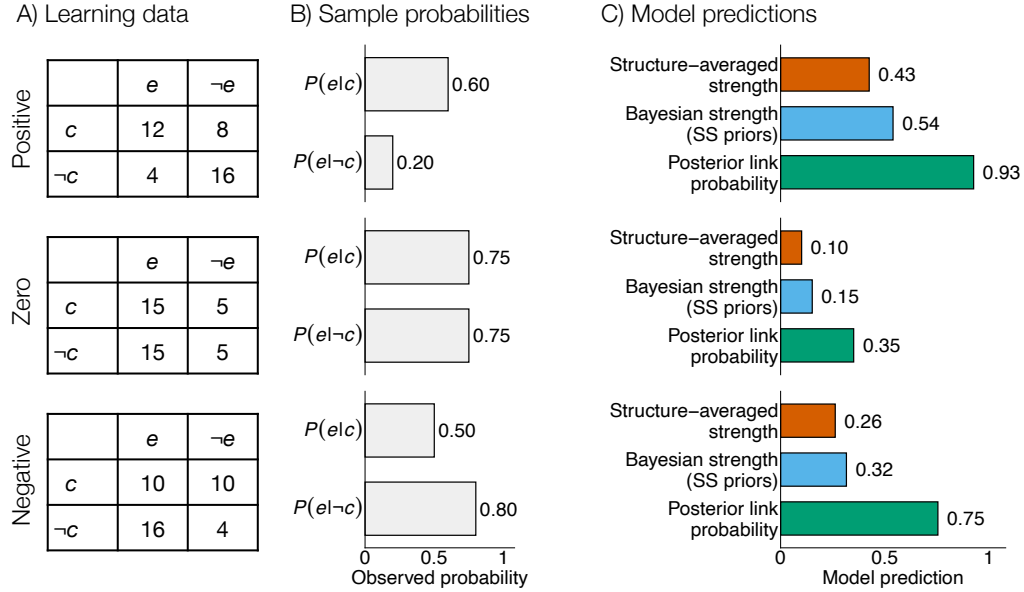


Figure 2

Example model predictions. A) Example scenarios for positive, zero, and negative contingencies between a cause C and an effect E . B) Observed probability of the effect given the presence and absence of the cause. C) Model predictions for the three scenarios, illustrating the distinct predictions of structure-averaged strength, Bayesian strength estimation under SS priors, and posterior link probability.

Structure-Averaged Strength Under Zero Contingency

Causal induction under zero contingency illustrates not only how and when the models' predictions differ, but also how different sources of uncertainty inform those predictions. Even though in this case the effect is equally likely in the presence and absence of

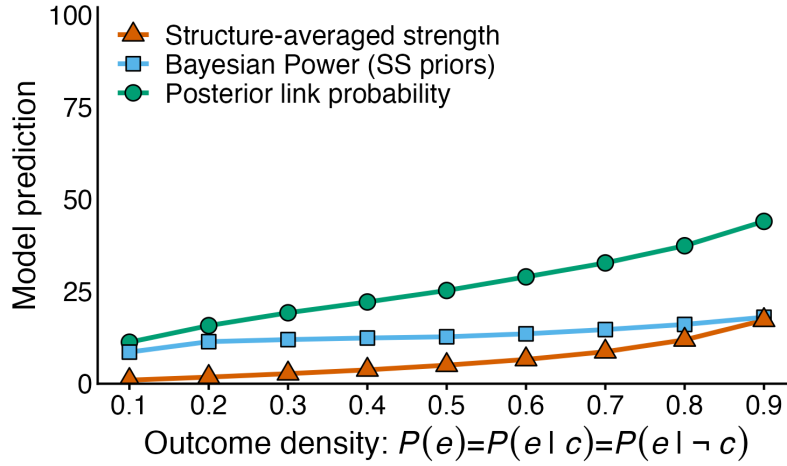
a candidate cause, $P(e | c) = P(e | \neg c)$ and hence $\Delta P = 0$, participants commonly report positive judgments of causality (Allan, 1993; Buehner et al., 2003; Chow et al., 2024; Jenkins & Ward, 1965; Kao & Wasserman, 1993; Wasserman et al., 1993). Because observing zero contingency is often taken to imply the absence of causality, these findings are often interpreted as cognitive bias and an *illusion of causality* (Blanco et al., 2026; Matute et al., 2015; Vinas et al., 2023). Moreover, the magnitude of these judgments varies systematically with outcome density: judgments increase as the effect becomes more frequent, even though contingency remains zero (Matute et al., 2022; Vallée-Tourangeau et al., 2005).

From a Bayesian perspective, however, neither positive judgments under zero contingency nor their increase with outcome density imply that people fall prey to a “causal illusion.” Because the absence of contingency in an observed sample often does not conclusively rule out a causal relation, the three Bayesian quantities can remain positive, albeit for different reasons. Figure 3 shows predictions for $N = 40$ zero-contingency observations, with the cause present in half of the cases (i.e., $P(c) = P(\neg c) = 0.5$) and outcome density varying from 0.1 to 0.9. Posterior link probability and structure-averaged strength use equal structure priors and uniform parameter priors, whereas conditional Bayesian strength is computed under the link structure S_1 with SS priors (Lu et al., 2008).

In contrast to contingency-based accounts, all three Bayesian models predict increasing judgments as outcome density rises, despite the observed contingency remaining zero. Intuitively, when the outcome is already frequent in the absence of the candidate cause (high outcome density), there are fewer cases in which an additional generative influence could become observable if it existed. In this case, even relatively large values of causal strength can remain compatible with little or no observed difference between cause-present and cause-absent cases. The models nevertheless differ substantially in their predictions because they reflect distinct inferences about the causal relation. Posterior link probability reflects residual belief that a causal relation exists, even if no contingency was observed. Bayesian strength reflects uncertainty about the strength of a possible influence under S_1 , together with generic assumptions about causal strengths (i.e., the SS prior). Finally, structure-averaged strength integrates both sources of uncertainty by weighting estimated strength under S_1 by the posterior probability that the link exists. Thus, qualitatively similar outcome-density effects can arise from different inferential mechanisms.

Structure-Averaged Strength and Human Judgments

The preceding analyses have shown how uncertainty about causal strength and causal structure can be incorporated into a normative account of strength estimation. We next consider the descriptive question of whether these different forms of uncertainty are reflected in human causal judgments. This question is particularly relevant because common learning tasks require participants to make causal judgments from limited evidence that may leave uncertainty about both the magnitude and the existence of a causal relation. Typically, learners are asked to rate the strength of a relation (e.g., on a scale from 0 to 100) and thus are not explicitly asked to judge whether a causal link exists. Nevertheless, their strength judgments may be sensitive to evidence about causal structure—not because learners report a structural quantity rather than a strength estimate, but because their estimates of causal strength may themselves reflect uncertainty about causal structure.

**Figure 3**

Model predictions as a function of outcome density under zero contingency. Structure-averaged strength, posterior link probability, and Bayesian Power under sparse-and-strong priors all predict increasing judgments as the outcome becomes more frequent but differ in magnitude. Predictions are based on $N = 40$ observations, with the cause present in 50% of cases and $P(e | c) = P(e | \neg c)$.

We examine this hypothesis by revisiting published studies of human causal judgment to assess the descriptive accuracy of structure-averaged strength, comparing the account’s predictions to posterior link probability and Bayesian causal power under SS priors. We began with the nine studies compiled by Perales and Shanks (2007), which have also been used to evaluate Bayesian strength under SS priors (Lu et al., 2008) and posterior link probability (Griffiths & Tenenbaum, 2009), and added further studies from the literature reporting numerical condition-level means (Buehner et al., 2003; Liljeholm & Cheng, 2009; White, 2003a, 2003b, 2004). The resulting dataset comprised 12 studies, 29 experiments, and 348 individual learning conditions, spanning positive, negative, and zero contingencies as well as diverse task framings, judgment questions, and learning procedures (Appendix B). Of these conditions, 53% had positive, 28% negative, and 19% zero contingency. The data set thus provides a heterogeneous yet broad basis for comparing the predictions of structure-averaged strength with human judgments.

From Model Predictions to Human Judgment

For each condition (observed event frequencies), we computed predictions from the three models and compared them with the reported mean human judgment (see Appendix B for details). Model predictions were derived in accordance with the causal framing of the task and mapped to a common scale ranging from -100 to $+100$, with positive values indicating a generative relation, negative values indicating a preventive relation, and zero indicating no causal influence.² Posterior link probability and structure-averaged strength

²The data from Perales and Shanks (2004) were available only through the compilation reported by Perales and Shanks (2007). As the study included three positive-contingency conditions and all mean judgments

were computed using equal prior probabilities over structures and uniform priors over model parameters, whereas Bayesian causal power was estimated under the link structure using sparse-and-strong priors.

Posterior link probability was derived from the posterior probabilities assigned to the competing causal structures. For generative and preventive conditions, the prediction was the posterior probability of the corresponding noisy-OR or noisy-AND-NOT link structure relative to the no-link structure. For tasks allowing either causal direction, we followed Griffiths and Tenenbaum (2009) and considered three hypotheses: a no-link structure, a generative link structure, and a preventive link structure. The prediction was given by the difference between the posterior probabilities of the two link structures, such that greater support for a generative relation yielded positive values and greater support for a preventive relation yielded negative values.

Predictions for Bayesian causal power under SS priors only consider the link structure using either a generative or inhibitory form. In generative settings, causal strength was estimated as the posterior mean under the noisy-OR link structure for generative settings and noisy-AND-NOT form for preventive framings. When the task allowed either direction, posterior mean strengths from the generative and preventive models were weighted by their relative marginal likelihoods and combined into a signed estimate, following Lu et al. (2008).

Structure-averaged causal strength additionally allowed for the possibility that no causal link existed. In tasks with a specified causal direction, the strength estimate under S_1 was weighted by the posterior probability of the corresponding generative or preventive link structure. In tasks that left causal direction open, strength was averaged over generative, preventive, and no-link structures, with each structure-specific estimate weighted by its posterior probability; the no-link structure contributed zero.

Model Comparison Across Studies

Figure 4 summarizes the models’ predictive accuracy across studies using two measures. Correlations indicate how well each model captures variation in judgments across conditions (Panel A), whereas root mean squared error (RMSE) measures the absolute size of prediction errors (Panel B).

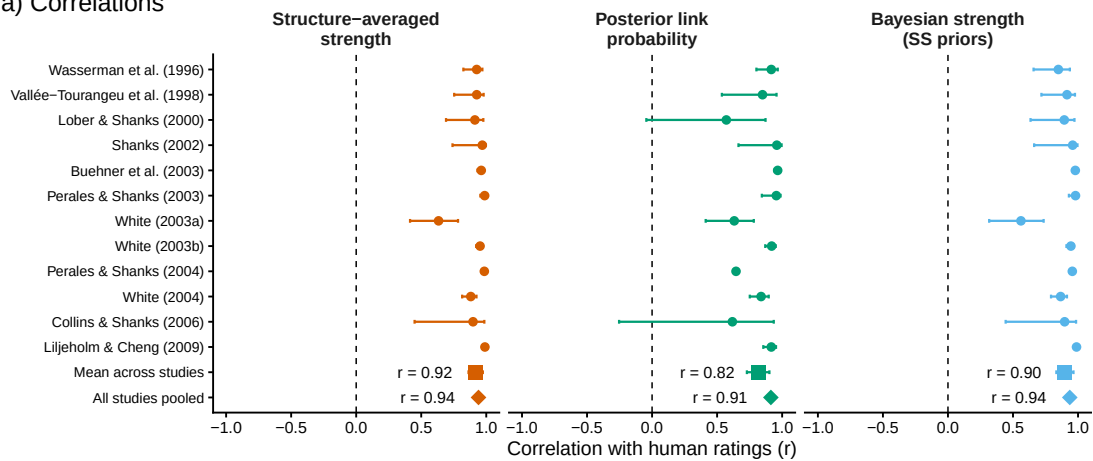
Structure-averaged causal strength best accounted for human judgments, yielding the highest average correlation and the lowest average RMSE. The differences between structure-averaged strength and Bayesian strength under SS priors were modest, however, suggesting that both strength-based models captured human judgments well. Posterior link probability performed less well, with lower correlations and substantially larger RMSEs across studies. The high predictive accuracy of structure-averaged strength is consistent with the idea that causal judgments reflect both estimated magnitude and uncertainty about link existence.

Figure 5 summarizes predictive accuracy separately for positive, negative, and zero-contingency conditions. For positive and negative contingencies, structure-averaged causal strength and Bayesian causal power under SS priors provided the best fits, with structure-averaged strength yielding the lowest prediction error in both cases. Interestingly, under zero contingency, posterior link probability captured judgments best, suggesting that that

were positive, we classified its framing as generative.

Model comparison

a) Correlations



b) Prediction error

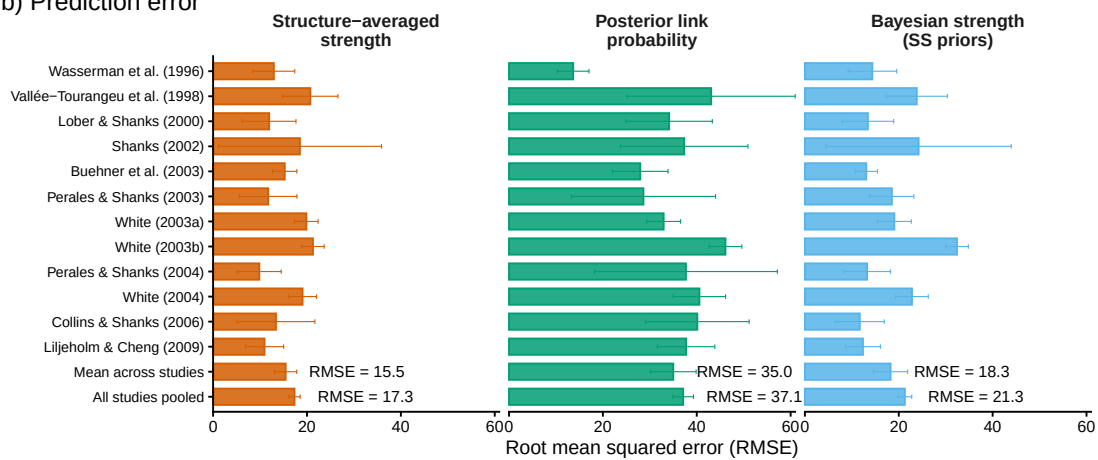


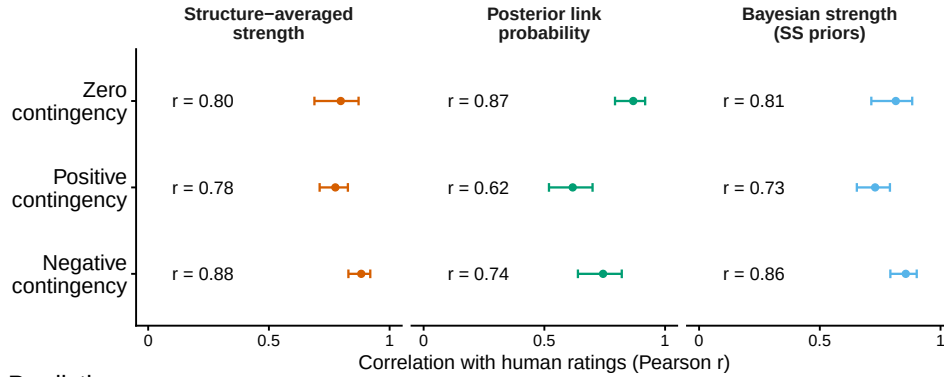
Figure 4

Model comparison across studies. a) Correlations between model predictions and human judgments (higher is better). b) Root mean squared error (lower is better). “Mean across studies” denotes the mean of the correlation and prediction error, respectively, computed separately for each of the 12 studies. The row labeled “All studies pooled” gives the correlation and error, respectively, computed for all 348 individual conditions pooled across studies.

evidence about whether a causal link exists may be particularly relevant when the observed contingency is zero and learners are uncertain about the existence of a causal relation (see also Stephan & Waldmann, 2026).

Model comparison by contingency type

a) Correlation



b) Prediction error

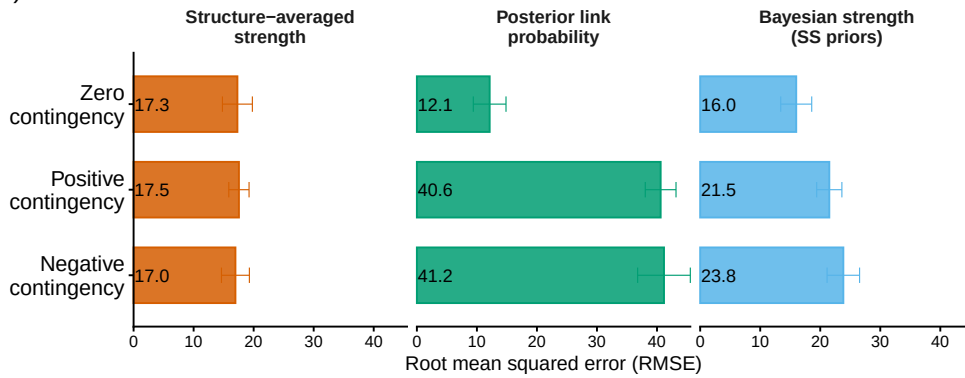


Figure 5

Model comparison by contingency type. a) Correlations between model predictions and human judgments (higher is better). b) Root mean squared error (lower is better). Contingency conditions were pooled across all studies.

We conducted three robustness checks. First, in a study-level bootstrap with 10,000 resamples, structure-averaged strength had the highest correlation and lowest RMSE in more than 99% of samples. Second, restricting the analysis to the nine studies compiled by Perales and Shanks (2007) produced the same ordering, although the advantage of structure-averaged strength over Bayesian causal power under SS priors was small (see also Lu et al., 2008). Third, we compared the three models with estimated strength under S_1 using uniform priors and structure-averaged strength using SS priors. Estimated strength under S_1 with uniform priors provided a slightly better numerical fit than structure-averaged strength with uniform priors, whereas structure-averaged strength with SS priors performed less well. This finding suggests that some participants may have approached the tasks with a strong prior expectation that a causal link was present (Ng et al., 2026) and focused primarily on estimating its strength.

Taken together, structure-averaged causal strength predicted human causal judgments well, suggesting that judgments of causal strength can reflect both causal magnitude and uncertainty about whether a causal relation exists. At the same time, posterior link probability provided the best account of zero-contingency judgments, which have historically motivated structure-based accounts (Griffiths & Tenenbaum, 2005). The additional analyses further indicate that Bayesian strength estimation under S_1 remains an important limiting case, with uniform priors providing a slightly better numerical fit than the corresponding structure-averaged estimate and the SS-prior version. Overall, this pattern indicates that judgments of causal strength can integrate structural and parametric information, with their relative contribution depending on the evidential context.

Discussion

What does it mean to estimate causal strength when the existence of the causal relation itself is uncertain? We addressed this question by proposing a Bayesian account in which uncertainty about whether a causal relation exists is incorporated into an estimate of its strength. Existing accounts have typically considered structural questions (whether a causal link exists) and strength questions (how strong a presumed link is) separately. We integrated these two forms of inference by defining structure-averaged strength as the normatively appropriate quantity when the inferential goal is to assess the strength of an uncertain causal relation. Unlike posterior link probability, this quantity retains causal magnitude as the inferential target. Unlike strength estimates derived under a single structure, it incorporates uncertainty about whether the link exists. The account therefore connects structural inference and strength estimation within a common normative framework.

The empirical analyses support the descriptive relevance of this distinction. Structure-averaged causal strength captured mean judgments well across a broad set of published studies, consistent with the idea that human strength judgments integrate information about both causal magnitude and uncertainty about the underlying structure. This shows that sensitivity to structural uncertainty does not require interpreting learners' responses as judgments about causal structure rather than strength. Instead, structural uncertainty may itself be reflected in the estimation of causal strength. Posterior link probability performed best under zero contingency, suggesting that the relative contribution of structural and parametric information may depend on the evidential context (cf. Stephan & Waldmann, 2026).

More broadly, the present work connects existing structure- and strength-based accounts within a common framework defined by two forms of uncertainty. Classic measures such as contingency and causal power provide point estimates under an assumed link structure and therefore correspond to the limiting case in which neither parameter nor structure uncertainty is represented (Cheng, 1997; Cheng & Novick, 1992). These measures are special cases within the Bayesian framework of causal induction. Causal power corresponds to the strength parameter in noisy-OR and noisy-AND-NOT causal models (Glymour, 2003), and both causal power and ΔP can be understood as maximum-likelihood estimates under their respective causal structures (Griffiths & Tenenbaum, 2005). Bayesian strength estimation under an assumed link represents parameter uncertainty while treating the relation as given (Lu et al., 2008; Yeung & Griffiths, 2015). The same Bayesian framework also supports inference about causal structure through quantities such as causal support

and posterior link probability, which represent uncertainty about whether a causal relation exists (Griffiths & Tenenbaum, 2005, 2009).

Structure-averaged causal strength brings these perspectives together by applying Bayesian model averaging to estimating causal strength, thereby integrating uncertainty about both causal magnitude and causal structure. This proposal thus aligns with and extends earlier applications of Bayesian model averaging to human causal learning and reasoning (Danks et al., 2002; Meder et al., 2014; Stephan & Waldmann, 2018). Normatively, our analyses clarify how causal strength should be estimated when the existence of the causal relation itself remains uncertain. Empirically, our findings suggest that human causal judgments reflect uncertainty about both causal strength and causal structure. Thus, our proposal connects structure and strength by showing how uncertainty about whether a causal relation exists can inform estimates of causal magnitude.

Transparency and Openness

All model code and data are publicly available at https://github.com/bmeder/causal_strength_structure_uncertainty

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Appendix A

Bayesian Inference Over Causal Structure and Strength

In Bayesian models of causal induction, the observed data can inform both which causal structure generated the data and what parameter values are likely within each structure. Posterior link probability and structure-averaged causal strength require inference about which causal structure generated the observed data. Structure-averaged strength additionally requires inference about causal strength conditional on each structure.

Causal structures and Bayesian learning

In the basic two-structure case, the hypothesis space contains two competing structures (Figure 1B). Structure S_1 states that there is a causal relation, such that effect E is a function of both the background causes A and cause C . The competing hypothesis, S_0 , represents the hypothesis that there is no causal relation: effect E occurs independently of the candidate cause C and is generated exclusively by the amalgam of background causes A .

Each structure is associated with a set of parameters, denoted as θ . For structure S_1 , the parameters include the base rate of the candidate cause, b_c , the causal strength w_c of C on E , and the influence w_a of the background cause A . Structure S_0 states that there is no causal relation between C and E and therefore includes only two parameters, the base rate of C , b_c and the strength of the background cause A , w_a . Note that the parameter b_c , representing the base rate of the candidate cause, is included for completeness but does not affect the comparison between causal structures because it is parameterized identically under the competing structures.

For computing posterior link probability and structure-averaged strength in the primary analyses, we use independent uniform Beta priors over the parameters of each structure (Griffiths & Tenenbaum, 2005; Meder et al., 2014; Yeung & Griffiths, 2015):

$$\begin{aligned} w_a &\sim \text{Beta}(1, 1), \\ b_c &\sim \text{Beta}(1, 1), \\ w_c &\sim \text{Beta}(1, 1) \quad \text{under } S_1. \end{aligned} \tag{A1}$$

These uniform priors represent an absence of specific prior assumptions about the parameter values. The Bayesian strength model used in the model comparison instead assumes the sparse-and-strong (SS) priors proposed by Lu et al. (2008), described below.

Using Bayes' theorem, the posterior probability distribution of each structure's parameters conditional on the observed data is given by

$$P(\theta \mid D; S_i) = \frac{P(D \mid \theta; S_i) P(\theta \mid S_i)}{P(D \mid S_i)}. \tag{A2}$$

where $P(D \mid \theta; S_i)$ is the likelihood of the data given the parameter values, $P(\theta \mid S_i)$ is the prior distribution over parameters, and $P(D \mid S_i)$ is the marginal likelihood of the data under structure S_i .

The posterior distribution represents the parameter values that remain plausible after observing the data. Of particular interest for the present analyses is the posterior distribution of causal strength, w_c , under a link structure. The posterior expectation, $\mathbb{E}[w_c \mid$

$D; S_1]$, provides the Bayesian estimate of causal strength conditional on that structure. Its numerical computation is described below.

Likelihood Computation

The likelihood function specifies how probable the observed data are for a given set of parameter values under a particular causal structure. We first specify the probability of the effect under each structure and then use these probabilities to construct the likelihood of the complete contingency table.

Effect probabilities. Under the link structure S_1 , the effect depends on both the candidate cause C and the background cause A , with the functional form depending on whether the relation is generative or preventive. For a generative link structure S_{1+} , we assume a noisy-OR parameterization (Glymour, 2003; Pearl, 1988). When the candidate cause is present, C and A provide independent opportunities to generate the effect:

$$P(e \mid c; \theta, S_{1+}) = w_c + w_a - w_c w_a. \quad (\text{A3})$$

Here, w_c represents the probability that the candidate cause generates an effect that would otherwise be absent, corresponding to the assumptions underlying causal power theory (Cheng, 1997).

For a preventive link structure S_{1-} , we use the corresponding noisy-AND-NOT parameterization (Griffiths & Tenenbaum, 2005; Yuille & Lu, 2007):

$$P(e \mid c; \theta, S_{1-}) = w_a(1 - w_c). \quad (\text{A4})$$

Here, w_c represents the probability that the candidate cause prevents an effect that would otherwise be generated by the background cause (Cheng, 1997).

When the candidate cause is absent, under both functional forms the effect probability is determined solely by the background cause:

$$P(e \mid \neg c; \theta, S_{1+}) = P(e \mid \neg c; \theta, S_{1-}) = w_a. \quad (\text{A5})$$

Under the no-link structure S_0 , the effect is independent of the candidate cause and is generated solely by the background cause. Thus, the causal strength parameter w_c is fixed to zero and the probability of the effect depends solely on the strength of the background cause:

$$P(e \mid c; \theta, S_0) = P(e \mid \neg c; \theta, S_0) = w_a. \quad (\text{A6})$$

Likelihood of the observed data. The effect probabilities above can be used to compute the likelihood of the data D , i.e., the complete contingency table, with each cell corresponding to one possible joint observation of cause and effect. The probability of each observation is determined by the base rate of the candidate cause, $b_c = P(c)$, together with the corresponding effect probability specified above. The likelihood is obtained by raising each of these four probabilities to its observed frequency and multiplying the four resulting terms.

Applying this procedure to the no-link structure S_0 , for which the effect probability is w_a regardless of whether C is present or absent, gives

$$\begin{aligned}
P(D \mid \theta; S_0) &= [(1 - b_c)(1 - w_a)]^{N(\neg c, \neg e)} \\
&\quad \times [(1 - b_c)w_a]^{N(\neg c, e)} \\
&\quad \times [b_c(1 - w_a)]^{N(c, \neg e)} \\
&\quad \times [b_c w_a]^{N(c, e)}.
\end{aligned} \tag{A7}$$

For the generative link structure S_{1+} , the first two terms remain unchanged (the cases in which C is absent), whereas the last two terms incorporate the noisy-OR relation specified in Equation A3:

$$\begin{aligned}
P(D \mid \theta; S_{1+}) &= [(1 - b_c)(1 - w_a)]^{N(\neg c, \neg e)} \\
&\quad \times [(1 - b_c)w_a]^{N(\neg c, e)} \\
&\quad \times [b_c(1 - w_c)(1 - w_a)]^{N(c, \neg e)} \\
&\quad \times [b_c(w_c + w_a - w_c w_a)]^{N(c, e)}.
\end{aligned} \tag{A8}$$

For the preventive link structure S_{1-} , the first two terms likewise remain unchanged, and the last two terms implement the noisy-AND-NOT relation specified in Equation A4:

$$\begin{aligned}
P(D \mid \theta; S_{1-}) &= [(1 - b_c)(1 - w_a)]^{N(\neg c, \neg e)} \\
&\quad \times [(1 - b_c)w_a]^{N(\neg c, e)} \\
&\quad \times [b_c(1 - w_a(1 - w_c))]^{N(c, \neg e)} \\
&\quad \times [b_c w_a(1 - w_c)]^{N(c, e)}.
\end{aligned} \tag{A9}$$

Equations A7, A8, and A9 thus express how well a given set of parameter values under each causal structure accounts for the observed contingency table. These likelihoods provide the basis for both parameter inference within each structure and inference over the alternative causal structures.

Marginal Likelihood

The likelihoods in Equations A7, A8, and A9 are conditional on fixed parameter values. Bayesian inference over structures additionally requires the marginal likelihood of the data under each structure, obtained by integrating the likelihood with respect to the prior distribution over the parameters.

For structure S_1 , the marginal likelihood is given by

$$P(D \mid S_1) = \iiint P(D \mid b_c, w_a, w_c; S_1) P(b_c, w_a, w_c \mid S_1) db_c dw_a dw_c, \tag{A10}$$

whereas for structure S_0 it is given by

$$P(D \mid S_0) = \iint P(D \mid b_c, w_a; S_0) P(b_c, w_a \mid S_0) db_c dw_a. \tag{A11}$$

Posterior Link Probability

The marginal likelihoods provide the basis for inferences about causal structure. Following Griffiths and Tenenbaum (2009), we use the posterior probability of the link structure, $P(S_1 | D)$, as the measure of structural evidence that a causal relation exists.

The posterior probability of each structure, $P(S_1 | D)$ and $P(S_0 | D)$, is computed by multiplying each structure’s marginal likelihood by its prior probability and then normalizing across structures:

$$P(S_i | D) = \frac{P(D | S_i)P(S_i)}{P(D)}, \quad (\text{A12})$$

with the normalizing constant $P(D)$ given by

$$P(D) = \sum_{i \in \{0,1\}} P(D | S_i)P(S_i) = P(D | S_0)P(S_0) + P(D | S_1)P(S_1). \quad (\text{A13})$$

In the two-structure case, the posterior link probability corresponds to the posterior probability of the link structure S_1 :

$$P(S_1 | D) = \frac{P(D | S_1)P(S_1)}{P(D | S_1)P(S_1) + P(D | S_0)P(S_0)}. \quad (\text{A14})$$

This quantity expresses the posterior probability that the observed data were generated by a structure containing a causal link from C to E , rather than by the no-link structure S_0 . Unless stated otherwise, we use uniform priors over the considered causal structures, i.e., $P(S_1) = P(S_0) = 0.5$ for the two-structure case and for the three-structure case involving both a generative and preventive structure $P(S_{1+}) = P(S_{1-}) = P(S_0) = 0.5$.

Structure-Averaged Causal Strength

Structure-averaged strength formalizes the idea that observed data can be informative in two ways: they can indicate how strong a causal relation would be if a link exists, and they can indicate how likely it is that such a link exists at all. Structure-averaged strength integrates both inferences by weighting the link-specific strength estimate by the posterior probability that a link exists.

Under structure S_0 , C and E are independent, and the strength of candidate cause C is fixed to zero:

$$w_c^{S_0} = 0. \quad (\text{A15})$$

Under S_1 , causal strength is represented by w_c , and is estimated by the posterior expectation

$$w_c^{S_1} = \mathbb{E}[w_c | D; S_1] = \int w_c P(w_c | D; S_1) dw_c. \quad (\text{A16})$$

To obtain a single strength estimate that accounts for uncertainty about whether a causal link exists at all, causal strength is averaged over structures. In the two-structure case, the resulting structure-averaged causal strength is

$$\begin{aligned}
\mathbb{E}[w_c \mid D] &= \sum_{i \in \{0,1\}} P(S_i \mid D) \mathbb{E}[w_c \mid D; S_i] \\
&= P(S_1 \mid D) \cdot w_c^{S_1} + P(S_0 \mid D) \cdot 0 \\
&= P(S_1 \mid D) \cdot w_c^{S_1}.
\end{aligned} \tag{A17}$$

Put simply, structure-averaged strength equals the estimated causal strength under S_1 multiplied by the posterior probability of this causal structure, i.e., the posterior link probability $P(S_1 \mid D)$. This estimate reflects both the inferred magnitude of the causal influence under S_1 and the posterior probability that a link exists.

The same model-averaging principle applies to larger hypothesis spaces. For example, if the hypothesis space contains a no-link structure S_0 , a generative link structure S_{1+} , and a preventive link structure S_{1-} , the signed structure-averaged estimate is

$$\begin{aligned}
\mathbb{E}[w_c \mid D] &= \sum_{i \in \{0,1+,1-\}} P(S_i \mid D) \mathbb{E}[w_c \mid D; S_i] \\
&= P(S_{1+} \mid D) w_c^{S_{1+}} - P(S_{1-} \mid D) w_c^{S_{1-}} + P(S_0 \mid D) \cdot 0 \\
&= P(S_{1+} \mid D) w_c^{S_{1+}} - P(S_{1-} \mid D) w_c^{S_{1-}}.
\end{aligned} \tag{A18}$$

Here, the no-link structure contributes zero strength, generative strength contributes positively, and preventive strength contributes negatively. The same principle can be applied to hypothesis spaces containing multiple candidate structures.

Bayesian Causal Strength Under Sparse-and-Strong Priors

The Bayesian estimate of causal strength conditional on the link structure depends on the prior distribution assigned to the strength parameters. Whereas posterior link probability and structure-averaged strength were computed using independent uniform parameter priors, the Bayesian strength model used in the model comparison assumes the sparse-and-strong (SS) priors of Lu et al. (2008).

The SS priors instantiate the idea of causal parsimony by favoring explanations in which relatively few causes are effective and effective causes tend to have substantial strength. In the notation used here, w_a denotes the strength of the background cause and w_c the strength of the candidate cause (corresponding to w_0 and w_1 , respectively, in Lu et al. (2008)). We use the SS strength prior defined for inference conditional on the link structure S_1 , rather than the augmented prior introduced for judgments about causal structure.

The sparse-and-strong (SS) prior changes the joint prior over the causal strength parameters w_c (strength of candidate cause) and w_a (strength of background cause), while the prior over the base rate of the candidate cause remains independent and uniform. The joint prior over (w_a, w_c) differs for generative and preventive relations because the two causal parameterizations imply different roles for the background and candidate causes.

Generative relations. For a noisy-OR (generative) relation, the background cause and candidate cause are both potential generators of the effect and therefore compete as alternative causal explanations. The SS prior favors configurations in which one cause is strong and the other is weak. Its density peaks at $(w_a, w_c) = (0, 1)$, where the candidate

cause is the sole strong cause, and at $(w_a, w_c) = (1, 0)$, where the background cause is the sole strong cause.

Following Equation 10 of Lu et al. (2008), the generative SS prior is given by

$$P(w_a, w_c \mid S_1, \text{gen}) \propto e^{-\alpha w_a - \alpha(1-w_c)} + e^{-\alpha(1-w_a) - \alpha w_c}. \quad (\text{A19})$$

The parameter α controls how strongly sparse-and-strong causal configurations are preferred. When $\alpha = 0$, the prior is uniform over w_a and w_c . As α increases, the prior becomes increasingly concentrated around $(0, 1)$ and $(1, 0)$. Following Lu et al. (2008), we use $\alpha = 5$.

Preventive relations. For a noisy-AND-NOT (preventive) relation, the background cause remains generative, whereas the candidate cause may prevent the effect. Consequently, the background and candidate causes do not compete as alternative generators. Because prevention can be expressed only when the effect would otherwise be generated, the prior favors a strong background cause, $w_a \approx 1$. It has one peak at $(w_a, w_c) = (1, 1)$, corresponding to a strong background cause and a strong preventive candidate cause, and another at $(w_a, w_c) = (1, 0)$, corresponding to a strong background cause and an ineffective candidate cause.

Following Equation 11 of Lu et al. (2008), the preventive SS prior is

$$P(w_a, w_c \mid S_1, \text{prev}) \propto e^{-\alpha(1-w_a) - \alpha(1-w_c)} + e^{-\alpha(1-w_a) - \alpha w_c}. \quad (\text{A20})$$

As in the generative case, $\alpha = 0$ yields a uniform prior, whereas larger values increasingly concentrate prior density around the two favored configurations. We again use $\alpha = 5$.

Posterior causal strength

The SS prior is combined with the appropriate likelihood under S_1 , using a noisy-OR parameterization for generative relations and a noisy-AND-NOT parameterization for preventive relations. The posterior distribution over the parameters is

$$P(\theta \mid D; S_1) = \frac{P(D \mid \theta; S_1) P(\theta \mid S_1)}{P(D \mid S_1)}. \quad (\text{A21})$$

The model prediction is the posterior mean of the candidate cause's strength conditional on S_1 :

$$\mathbb{E}[w_c \mid D; S_1] = \int w_c P(\theta \mid D; S_1) dw_c. \quad (\text{A22})$$

For a generative relation, w_c represents the probability that the candidate cause generates an effect that would otherwise be absent. For a preventive relation, w_c represents the probability that the candidate cause prevents an effect that would otherwise occur.

Model implementation

All model and analysis code is available at https://github.com/bmeder/causal_strength_structure_uncertainty. The R code uses Monte Carlo simulation to approximate the integrals by drawing m independent samples from the respective parameter priors.

For posterior link probability and structure-averaged strength in the primary model comparison, samples were drawn from independent uniform Beta(1, 1) parameter priors. Under a link structure S_1 , each sample consists of $\theta^{(k)} = (b_c^{(k)}, w_c^{(k)}, w_a^{(k)})$, with all three parameters drawn independently from uniform Beta(1, 1) distributions. In the three-structure case, the same parameterization applies separately to the generative and preventive link structures, S_{1+} and S_{1-} , using their respective likelihood functions defined above. Under S_0 , the structure parameters are b_c and w_a , with w_c fixed to zero.

For Bayesian causal strength under the sparse-and-strong prior, $(w_a^{(k)}, w_c^{(k)})$ was instead drawn from the corresponding generative or preventive SS prior (Equations A19 and A20), while b_c was drawn independently from a uniform Beta(1, 1) prior.

For structure inference, the marginal likelihood of each structure is approximated by averaging the likelihood over prior samples:

$$P(D | S_i) \approx \frac{1}{m} \sum_{k=1}^m P(D | \theta^{(k)}; S_i). \quad (\text{A23})$$

where each $\theta^{(k)}$ is one sample from the parameter prior.

In the three-structure case, the same parameterization applies separately to the generative and preventive link structures, S_{1+} and S_{1-} , using their respective likelihood functions defined above.

The posterior expectation of causal strength under the link structure S_1 is computed by likelihood-weighted averaging over the same prior samples:

$$\mathbb{E}[w_c | D; S_1] \approx \frac{\sum_{k=1}^m w_c^{(k)} P(D | \theta^{(k)}; S_1)}{\sum_{k=1}^m P(D | \theta^{(k)}; S_1)}. \quad (\text{A24})$$

For Bayesian causal strength under SS priors, this posterior expectation under S_1 constitutes the model prediction and is computed using the SS prior.

For structure-averaged strength, the corresponding posterior expectation computed under uniform parameter priors is additionally weighted by the posterior probability of the link structure. In the two-structure case, since $w_c = 0$ under S_0 , this reduces to

$$\mathbb{E}[w_c | D] = P(S_1 | D) \mathbb{E}[w_c | D; S_1]. \quad (\text{A25})$$

For the three-structure case, structure-averaged strength is computed analogously from the posterior probabilities and posterior mean strengths of S_{1+} and S_{1-} (cf. Equation A18).

Appendix B

Model Comparison

Table B1 provides an overview of the studies included in the model comparison. The full data set with additional information, including observed event frequencies and human judgments, is provided online at https://github.com/bmeder/causal_strength_structure_uncertainty.

Table B1

Overview of studies included in the model comparison

Study	Experiments	Causal framing	Trials	Conditions
Vallée-Tourangeau et al. (1998)	1–2	Bidirectional	40	12
Wasserman et al. (1996)	1	Bidirectional	20–36	21
Lober and Shanks (2000)	1–3	Generative	56–60	11
Shanks (2002)	1–2	Bidirectional	60	6
Perales and Shanks (2003)	1–3	Bidirectional	20	12
White (2003a)	1–2	Bidirectional	12–30	64
White (2003b)	1–6	Generative	40–48	44
Buehner et al. (2003)	1–3	Gen./ Prev.	16–72	50
White (2004)	1–2	Bidirectional	40	70
Collins and Shanks (2006)	1–2	Generative	80–90	7
Perales and Shanks (2004) ^a	1a	Generative	80	3
Liljeholm and Cheng (2009)	1–2	Gen./ Prev.	32–128	48

Note. Number of learning trials are shown as ranges when they varied across experiments or conditions within a study. Causal framing indicates whether the tasks specified a generative relation, a preventive relation, or allowed both causal directions (bidirectional).

^a Study information and data were obtained from Perales and Shanks (2007), as the original source was not available.

To evaluate how well the models account for human judgments, each model’s predictions were generated using the observed cell frequencies from each contingency condition in each experiment. Model predictions were derived in accordance with the causal framing of the task and mapped to the common response scale. Generative predictions were multiplied by 100, preventive predictions by -100 . Bidirectional predictions for conditions in which judgments were given on a -100 to $+100$ scale were computed as signed combinations of generative and preventive components.

Structure-averaged causal strength

For structure-averaged strength, predictions incorporated uncertainty about whether a link existed. For generative and preventive tasks, predictions were

$$\hat{y} = 100 \cdot P(S_{1+} \mid D) \cdot \mathbb{E}[w_c \mid D; S_{1+}] \quad (\text{B1})$$

and

$$\hat{y} = -100 \cdot P(S_{1-} \mid D) \cdot \mathbb{E}[w_c \mid D; S_{1-}], \quad (\text{B2})$$

For bidirectional tasks, the hypothesis space contained S_0 , S_{1+} , and S_{1-} , with equal structure priors. The signed prediction was

$$\begin{aligned}\hat{y} = & 100 \cdot P(S_{1+} \mid D) \cdot \mathbb{E}[w_c \mid D; S_{1+}] \\ & - 100 \cdot P(S_{1-} \mid D) \cdot \mathbb{E}[w_c \mid D; S_{1-}] \\ & + 100 \cdot P(S_0 \mid D) \cdot 0.\end{aligned}\tag{B3}$$

Thus, generative strength contributed positively, preventive strength negatively, and the no-link structure contributed zero.

Posterior link probability

Posterior link probability was used as a measure of structural evidence, with predictions generated following Griffiths and Tenenbaum (2009). For generative and preventive tasks, predictions were the posterior probability of the corresponding link structure relative to the no-link structure. For generative task framing, model prediction were given by

$$\hat{y} = 100 \cdot P(S_{1+} \mid D).\tag{B4}$$

For preventive task framing, model prediction were given by

$$\hat{y} = -100 \cdot P(S_{1-} \mid D).\tag{B5}$$

For bidirectional tasks, the hypothesis space contained three possible structures: a no-link structure, a generative-link structure, and a preventive-link structure. These structures were assigned equal prior probabilities. Predictions were mapped to the signed response scale by treating posterior probability assigned to a generative link as positive and posterior probability assigned to a preventive link as negative (Griffiths & Tenenbaum, 2009, p. 681):

$$\hat{y} = 100 \cdot P(S_{1+} \mid D) - 100 \cdot P(S_{1-} \mid D).\tag{B6}$$

Bayesian causal power under sparse-and-strong (SS) priors

For the Bayesian causal-power model with SS priors (Lu et al., 2008), predictions were the posterior mean causal strength conditional on the generative or preventive link structure S_1 , depending on task framing. For generative framing, predictions were

$$\hat{y} = 100 \cdot \mathbb{E}[w_c \mid D; S_{1+}]\tag{B7}$$

and for preventive framing

$$\hat{y} = -100 \cdot \mathbb{E}[w_c \mid D; S_{1-}],\tag{B8}$$

For bidirectional tasks, following Equation 15 of Lu et al. (2008), the posterior mean strengths under the generative and preventive link structures were combined as signed quantities, weighted by the posterior probabilities of the two causal directions conditional on a link being present:

$$\begin{aligned}\hat{y} = & 100 \cdot P(S_{1+} \mid D, S_1) \cdot \mathbb{E}[w_c \mid D; S_{1+}] \\ & - 100 \cdot P(S_{1-} \mid D, S_1) \cdot \mathbb{E}[w_c \mid D; S_{1-}],\end{aligned}\tag{B9}$$

where

$$P(S_{1+} \mid D, S_1) = \frac{P(D \mid S_{1+})}{P(D \mid S_{1+}) + P(D \mid S_{1-})}\tag{B10}$$

and analogously for $P(S_{1-} \mid D, S_1)$.